



CourseCube®
(Formerly Java Learning Center)

MicroServices

Study Guide

Module 3

Developing BookStore –V.1.1
(Actuators and Boot Admin)

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Developing BookStore V-1.1

Applying Actuators in our BookStore App Eco-System

- Actuator brings production-ready features to application without implementing these features in the application.
- When application is in Production , you need to check the following
** Health Check , Auditing , Metrics gathering , Monitoring etc **

Step A : Using Actuators in BookPriceMS

BookPriceMS: Port – 9000

1) Add Actuators related dependency to pom.xml of BookPriceMS:

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

2) Update pom.xml with the following

```
<version>1.1</version>
...
...
<description>This is BookPrice MS developed by Srinivas Dande</description>
```

3) Add Actuator related properties to application.properties of BookPriceMS:

```
management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

4) Start One Instance of BookPriceMS



5) Check the Actuator Endpoints of One Instance

<http://localhost:9000/actuator>

Step B : Using Actuators in BookSearchMS

BookSearchMS: Port – 8000

1) Add Actuators related dependency to pom.xml of BookSearchMS:

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

2) Update pom.xml with the following

```
<version>1.1</version>
...
...
<description>This is BookSearchMS MS developed by Srinivas Dande</description>
```

3) Add Actuator related properties to application.properties of BookSearchMS:

```
management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

4) Start One Instance of BookSearchMS

5) Check the Actuator Endpoints of One Instance

<http://localhost:8000/actuator>



Step C : Using Actuators in PlaceOrderMS

PlaceOrderMS: Port – 7000

1) Add Actuators related dependency to pom.xml of PlaceOrderMS:

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

2) Update pom.xml with the following

```
<version>1.1</version>
...
...
<description>This is PlaceOrderMS developed by Srinivas Dande</description>
```

3) Add Actuator related properties to application.properties of PlaceOrderMS:

```
management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

4) Start One Instance of PlaceOrderMS

5) Check the Actuator Endpoints of One Instance

<http://localhost:7000/actuator>



Step D : Using Actuators in UserRatingMS

UserRatingMS Port – 6500

1) Add Actuators related dependency to pom.xml of UserRatingMS:

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

2) Update pom.xml with the following

```
<version>1.1</version>
...
...
<description>This is UserRatingMS developed by Srinivas Dande</description>
```

3) Add Actuator related properties to application.properties of UserRatingMS:

```
management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

4) Start One Instance of UserRatingMS

5) Check the Actuator Endpoints of One Instance

<http://localhost:6500/actuator>

Step E : Using Actuators in BookStoreWeb

BookStoreWeb Port – 5000

1) Add Actuators related dependency to pom.xml of BookStoreWeb:

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

2) Update pom.xml with the following

```
<version>1.1</version>
...
...
<description>This is BookStoreWeb developed by Srinivas Dande</description>
```

3) Add Actuator related properties to application.properties of BookStoreWeb:

```
management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

4) Start One Instance of BookStoreWeb

5) Check the Actuator Endpoints of One Instance

<http://localhost:5000/actuator>

Spring Boot Admin Server

- Spring Boot Admin is not a core module provided by the Spring team, it was created by a company called Codecentric. Still, the code is publicly available on Github and it is free.
- Spring Boot Admin actually comes in two parts –
 - Boot Admin Server
 - Boot Admin Client
- The Server part contains the Admin user interface and runs independently from the monitored applications.
- The Client part is in the monitored application, which registers with the Admin Server part.

Setup Boot Admin Server:

<https://start.spring.io/> --- Spring Boot initializer portal

1) Create a Spring boot project from Spring Boot initializer portal with following

- Project** : Maven
- Language** : Java
- Spring Boot** : 2.5.3
- Project Metadata**
 - Group** : com.jlcindia.mybootadminserver
 - Artifact** : MyBootAdminServer
 - Name** : MyBootAdminServer
 - Description** : This is Boot Admin Server Written By Srinivas Dande
 - Package name** : com.jlcindia.mybootadminserver
 - Packaging** : jar
 - Java** : 8
- Dependencies**
 - Boot Admin Server
 - Spring Boot Actuator

and Click on Generate to download the project.

2) Unzip and import the project into Eclipse as existing maven project.

3) Delete src/test/java

4) Update the pom.xml



5) Now open **MyBootAdminServerApplication** class and add **@EnableAdminServer** annotation on the class.

```
@SpringBootApplication
@EnableAdminServer
public class MyBootAdminServerApplication {

    public static void main(String[] args) {
        SpringApplication.run(MyBootAdminServerApplication.class, args);
    }
}
```

6) **Update application.properties**

```
spring.application.name=MyBootAdminServer
server.port=5555
```

7) **Start Boot Admin Server**

8) **Test Boot Admin Server**

Open browser and go to **http://localhost:5555/**

Boot Admin Server: Files required

1. MyBootAdminServerApplication.java	2. application.properties
3. pom.xml	

1. MyBootAdminServerApplication.java

```
package com.jlcindia.mybootadminserver;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

import de.codecentric.boot.admin.server.config.EnableAdminServer;

@SpringBootApplication
@EnableAdminServer
public class MyBootAdminServerApplication {

    public static void main(String[] args) {
        SpringApplication.run(MyBootAdminServerApplication.class, args);
    }
}
```



2. application.properties

spring.application.name=MyBootAdminServer

server.port=5555

3. pom.xml

<project ...>

<groupId>**com.jlcindia.mybootadminserver**</groupId>

<artifactId>**MyBootAdminServer**</artifactId>

<version>**1.1**</version>

<name>**MyBootAdminServer**</name>

<description>This is Boot Admin Server Written By Srinivas Dande</description>

<properties>

<java.version>1.8</java.version>

<**spring-boot-admin.version**>2.4.3</spring-boot-admin.version>

</properties>

<parent>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-parent</artifactId>

<**version**>2.5.3</version>

<relativePath/>

</parent>

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<**artifactId**>spring-boot-starter-actuator</artifactId>

</dependency>

<dependency>

<groupId>de.codecentric</groupId>

<**artifactId**>spring-boot-admin-starter-server</artifactId>

</dependency>

</dependencies>



```
<dependencyManagement>
<dependencies>
<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-dependencies</artifactId>
<version>${spring-boot-admin.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```

Step A : Adding Boot Admin Client in BookPriceMS

1) Add Boot Admin Client related dependency to pom.xml of BookPriceMS

```
<dependency>
  <groupId>de.codecentric</groupId>
  <artifactId>spring-boot-admin-starter-client</artifactId>
  <version>2.4.3</version>
</dependency>
```

2) Add Boot Admin Client related properties to application.properties of BookPriceMS

```
spring.boot.admin.client.url=http://localhost:5555
```

3) Start One Instance of BookPriceMS

4) Check whether BookPriceMS Instance is registered in the BootAdmin Server.



Step B : Adding Boot Admin Client in BookSearchMS

1) Add Boot Admin Client related dependency to pom.xml of BookSearchMS

```
<dependency>
  <groupId>de.codecentric</groupId>
  <artifactId>spring-boot-admin-starter-client</artifactId>
  <version>2.4.3</version>
</dependency>
```

2) Add Boot Admin Client related properties to application.properties of BookSearchMS

```
spring.boot.admin.client.url=http://localhost:5555
```

3) Start One Instance of BookSearchMS

4) Check whether BookSearchMS Instance is registered in the BootAdmin Server.

Step C : Adding Boot Admin Client in PlaceOrderMS

1) Add Boot Admin Client related dependency to pom.xml of PlaceOrderMS

```
<dependency>
  <groupId>de.codecentric</groupId>
  <artifactId>spring-boot-admin-starter-client</artifactId>
  <version>2.4.3</version>
</dependency>
```

2) Add Boot Admin Client related properties to application.properties of PlaceOrderMS

```
spring.boot.admin.client.url=http://localhost:5555
```

3) Start One Instance of PlaceOrderMS

4) Check whether PlaceOrderMS Instance is registered in the BootAdmin Server.



Step D : Adding Boot Admin Client in UserRatingMS

1) Add Boot Admin Client related dependency to pom.xml of UserRatingMS

```
<dependency>  
  <groupId>de.codecentric</groupId>  
  <artifactId>spring-boot-admin-starter-client</artifactId>  
  <version>2.4.3</version>  
</dependency>
```

2) Add Boot Admin Client related properties to application.properties of UserRatingMS

```
spring.boot.admin.client.url=http://localhost:5555
```

3) Start One Instance of UserRatingMS

4) Check whether UserRatingMS Instance is registered in the BootAdmin Server.

Step E : Adding Boot Admin Client in BookStoreWeb

1) Add Boot Admin Client related dependency to pom.xml of BookStoreWeb

```
<dependency>  
  <groupId>de.codecentric</groupId>  
  <artifactId>spring-boot-admin-starter-client</artifactId>  
  <version>2.4.3</version>  
</dependency>
```

2) Add Boot Admin Client related properties to application.properties of BookStoreWeb

```
spring.boot.admin.client.url=http://localhost:5555
```

3) Start One Instance of BookStoreWeb

4) Check whether BookStoreWeb Instance is registered in the BootAdmin Server.



BookPriceMS – V-1.1: Files required

1. BookPriceController.java	Same as V.1.0
2. BookPriceService.java	Same as V.1.0
3. BookPriceServiceImpl.java	Same as V.1.0
4. BookPriceDAO.java	Same as V.1.0
5. BookPrice.java	Same as V.1.0
6. JLCBookPriceConfig.java	Same as V.1.0
7. BookPriceApplication.java	Same as V.1.0
8. application.properties	Updated in V1.1
9. pom.xml	Updated in V1.1

8. application.properties

spring.application.name=**MyBookPriceMS**

server.port=**9000**

spring.boot.admin.client.url=**http://localhost:5555**

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcbookpricedb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update

INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@



info.app.developer=Srinivas Dande
info.app.company=Java Learning Center

9. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.bookprice</groupId>
<artifactId>BookPriceMS</artifactId>
<version>1.1</version>
<packaging>jar</packaging>

<name>MyBookPriceMS</name>
<url>https://www.jlcindia.com</url>
<description>This is BookPrice MS developed by Srinivas Dande</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```



```
<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>MyBookPriceMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```



BookSearchMS – V-1.1: Files required

1. BookController.java	Same as V1.0
2. BookPriceInfo.java	Same as V1.0
3. BookInfo.java	Same as V1.0
4. BookService.java	Same as V1.0
5. BookServiceImpl.java	Same as V1.0
6. BookDAO.java	Same as V1.0
7. BookRatingDAO.java	Same as V1.0
8. BookInventoryDAO.java	Same as V1.0
9. Book.java	Same as V1.0
10. BookRating.java	Same as V1.0
11. BookInventory.java	Same as V1.0
12. JLCBookSearchConfig.java	Same as V1.0
13. BookSearchApplication.java	Same as V1.0
14. application.properties	Updated in V1.1
15. pom.xml	Updated in V1.1

14. application.properties

spring.application.name=**MyBookSearchMS**

server.port=**8000**

spring.boot.admin.client.url=**http://localhost:5555**

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcbooksdb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update



```
# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

15. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.booksearch</groupId>
<artifactId>BookSearchMS</artifactId>
<version>1.1</version>
<packaging>jar</packaging>

<name>MyBookSearchMS</name>
<url>https://www.jlcindia.com</url>
<description>This is BookSeach MS developed by Srinivas Dande</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>MyBookSearchMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```



PlaceOrderMS – V-1.1: Files required

1. OrderController.java	Same as V1.0
2. OrderInfo.java	Same as V1.0
3. OrderService.java	Same as V1.0
4. OrderServiceImpl.java	Same as V1.0
5. OrderDAO.java	Same as V1.0
6. OrderItemDAO.java	Same as V1.0
7. BookInventoryDAO.java	Same as V1.0
8. Order.java	Same as V1.0
9. OrderItem.java	Same as V1.0
10. BookInventory.java	Same as V1.0
11. JLCPlaceOrderConfig.java	Same as V1.0
12. PlaceOrderApplication.java	Same as V1.0
13.application.properties	Updated in V1.1
14.pom.xml	Updated in V1.1

13. application.properties

spring.application.name=**MyPlaceOrderMS**

server.port=**7000**

spring.boot.admin.client.url=**http://localhost:5555**

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcordersdb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update



```
# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

14. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.placeorder</groupId>
<artifactId>PlaceOrderMS</artifactId>
<version>1.1</version>
<packaging>jar</packaging>

<name>MyPlaceOrderMS</name>
<url>https://www.jlcindia.com</url>
<description>This is PlaceOrderMS developed by Srinivas Dande</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>MyPlaceOrderMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```



UserRatingMS – V-1.1: Files required

1. RatingController.java	Same as V1.0
2. RatingService.java	Same as V1.0
3. RatingServiceImpl.java	Same as V1.0
4. UserRatingDAO.java	Same as V1.0
5. BookRatingDAO.java	Same as V1.0
6. UserRating.java	Same as V1.0
7. BookRating.java	Same as V1.0
8. JLCUserRatingConfig.java	Same as V1.0
9. UserRatingApplication.java	Same as V1.0
10.application.properties	Updated in V1.1
11.pom.xml	Updated in V1.1

10. application.properties

spring.application.name=**MyUserRatingMS**

server.port=**6500**

spring.boot.admin.client.url=**http://localhost:5555**

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcratingsdb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update



INFO ENDPOINT CONFIGURATION

```
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

11. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.userrating</groupId>
<artifactId>UserRatingMS</artifactId>
<version>1.1</version>
<packaging>jar</packaging>

<name>MyUserRatingMS</name>
<url>https://www.jlcindia.com</url>
<description>This is UserRatingMS developed by Srinivas Dande</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>
</dependencies>

<build>
<finalName>MyUserRatingMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



BookStoreWeb – V-1.1: Files required

1. index.jsp	Same as V1.0
2. myheader.jsp	Same as V1.0
3. myfooter.jsp	Same as V1.0
4. showBooksList.jsp	Same as V1.0
5. showBookInfo.jsp	Same as V1.0
6. showCart.jsp	Same as V1.0
7. orderSuccess.jsp	Same as V1.0
8. addRating.jsp	Same as V1.0
9. ratingSuccess.jsp	Same as V1.0
10. ratingsList.jsp	Same as V1.0
11. bookstore.css	Same as V1.0
12. Book.java	Same as V1.0
13. BookInfo.java	Same as V1.0
14. Order.java	Same as V1.0
15. OrderItem.java	Same as V1.0
16. OrderInfo.java	Same as V1.0
17. UserRating.java	Same as V1.0
18. BookStoreController.java	Same as V1.0
19. BookStoreService.java	Same as V1.0
20. BookStoreServiceImpl.java	Same as V1.0
21. JLCBookStoreWebConfig.java	Same as V1.0
22. BookStoreWebApplication.java	Same as V1.0
23.application.properties	Updated in V1.1
24.pom.xml	Updated in V1.1

23. application.properties

spring.application.name=**MyBookStoreWeb**

server.port=**5000**

spring.boot.admin.client.url=**http://localhost:5555**

management.endpoints.web.exposure.include=*****
management.endpoint.health.show-details=**always**



```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

```
# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

24.pom.xml

```
<project .....>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.bookstoreweb</groupId>
<artifactId>BookStoreWeb</artifactId>
<version>1.1</version>
<packaging>jar</packaging>

<name>MyBookStoreWeb</name>
<url>https://www.jlcindia.com</url>
<description>This is BookStoreWeb developed by Srinivas Dande</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
```



```
<build>
<finalName>MyBookStoreWeb</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```